Data structures

Homework :

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1. (a) Define an integer array. Declare an integer pointer and initialize it to point to the first element of the array. Use pointer arithmetic to traverse the array and print the elements.

```
#include <stdio.h>

int main()
{
   int arr[5] = {10, 20, 30, 40, 50};
   int *p = arr;
   int i;

   printf("Array elements:\n");

   for(i = 0; i < 5; i++)
   {
      printf("%d ", *(p + i));
   }

   return 0;
}
```

Output :

```
Array elements:
10 20 30 40 50

=== Code Execution Successful ===
```

(b) Modify the elements of the array using pointer arithmetic.

```
#include <stdio.h>

int main()
{
   int arr[5] = {10, 20, 30, 40, 50};
```

```
    int *p = arr;
    int i;

    for(i = 0; i < 5; i++)
    {
        *(p + i) = *(p + i) + 5;
    }

    printf("Modified array:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d ", *(p + i));
    }

    return 0;
}
```

Output :

```
Modified array:
15 25 35 45 55

=== Code Execution Successful ===
```

2. Calculate string length using pointers.

```
#include <stdio.h>

int main()
{
    char str[100];
    char *p;
    int length = 0;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    p = str;

    while(*p != '\0')
    {
        length++;
```

```
    p++;
  }

  printf("Length = %d", length);

  return 0;
}
```

Output :

```
Enter a string: sripriyan
Length = 9

=== Code Execution Successful ===
```

3. Write a function that calculates both the sum and product of two numbers. Return these two results using pointers passed as arguments to the function.

```
#include <stdio.h>

void calculate(int a, int b, int *sum, int *product)
{
  *sum = a + b;
  *product = a * b;
}

int main()
{
  int a, b;
  int sum, product;

  printf("Enter two numbers: ");
  scanf("%d %d", &a, &b);

  calculate(a, b, &sum, &product);

  printf("Sum = %d\n", sum);
  printf("Product = %d", product);

  return 0;
}
```

Output :

```
Enter two numbers: 15
155
Sum = 170
Product = 2325

=== Code Execution Successful ===
```

4. Create an array of structures. Use dynamic memory allocation to allocate memory space and use a structure pointer with pointer arithmetic to iterate and access / modify members.

```
#include <stdio.h>
#include <stdlib.h>

struct Student
{
    int id;
    char name[30];
    float grade;
};

int main()
{
    struct Student *s;
    int n, i;

    printf("Enter number of students: ");
    scanf("%d", &n);

    s = (struct Student *)malloc(n * sizeof(struct Student));

    if(s == NULL)
    {
        printf("Memory allocation failed");
        return 1;
    }

    for(i = 0; i < n; i++)
    {
        printf("\nEnter details of student %d:\n", i + 1);
```

```
        printf("ID: ");
        scanf("%d", &(s + i)->id);

        printf("Name: ");
        scanf("%s", (s + i)->name);

        printf("Grade: ");
        scanf("%f", &(s + i)->grade);
    }

    printf("\nStudent Details:\n");

    for(i = 0; i < n; i++)
    {
        printf("\nID: %d", (s + i)->id);
        printf("\nName: %s", (s + i)->name);
        printf("\nGrade: %.2f\n", (s + i)->grade);
    }

    free(s);

    return 0;
}
```

Output :

```
Enter number of students: 2

Enter details of student 1:
ID: 2025115020
Name: sripriyan
Grade: 80

Enter details of student 2:
ID: 2025225000
Name: ace
Grade: 81

Student Details:

ID: 2025115020
Name: sripriyan
Grade: 80.00

ID: 2025225000
```

```
Name: ace
Grade: 81.00
```

```
=== Code Execution Successful ===
```

5. Define two functions with the same signature. Declare a function pointer and use the pointer to call both functions

```
#include <stdio.h>

void add(int a, int b)
{
   printf("Addition = %d\n", a + b);
}

void multiply(int a, int b)
{
   printf("Multiplication = %d\n", a * b);
}

int main()
{
   void (*fp)(int, int);

   fp = add;
   fp(10, 5);

   fp = multiply;
   fp(10, 5);

   return 0;
}
```

Output :

```
Addition = 15
Multiplication = 50


=== Code Execution Successful ===
```

6. Declare an integer, a pointer to it, and a double pointer. Print the values and addresses at each level.

```
#include <stdio.h>

int main()
{
   int x = 10;
   int *p = &x;
   int **q = &p;

   printf("Value of x = %d\n", x);
   printf("Address of x = %p\n", (void*)&x);

   printf("\nValue of p = %p\n", (void*)p);
   printf("Address of p = %p\n", (void*)&p);
   printf("Value pointed by p = %d\n", *p);

   printf("\nValue of q = %p\n", (void*)q);
   printf("Address of q = %p\n", (void*)&q);
   printf("Value pointed by q = %p\n", (void*)*q);
   printf("Value of x using **q = %d\n", **q);

   return 0;
}
```

Output :

```
Value of x = 10
Address of x = 0x7ffdbcf0fe6c

Value of p = 0x7ffdbcf0fe6c
Address of p = 0x7ffdbcf0fe60
Value pointed by p = 10
```

```
Value of q = 0x7ffdbcf0fe60
Address of q = 0x7ffdbcf0fe58
Value pointed by q = 0x7ffdbcf0fe6c
Value of x using **q = 10
```

```
=== Code Execution Successful ===
```

7. Sort an array of floating point numbers using Selection sort.

```c
#include <stdio.h>

int main()
{
    float arr[5] = {4.5, 1.2, 8.7, 2.3, 6.1};
    float temp;
    int i, j, min;

    for(i = 0; i < 4; i++)
    {
        min = i;

        for(j = i + 1; j < 5; j++)
        {
            if(arr[j] < arr[min])
            {
                min = j;
            }
        }

        temp = arr[i];
        arr[i] = arr[min];
        arr[min] = temp;
    }

    printf("Sorted array:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%.1f ", arr[i]);
    }

    return 0;
}
```

Output :

```
Sorted array:
1.2 2.3 4.5 6.1 8.7

=== Code Execution Successful ===
```

8. Define a structure Student with id, name, and grade. Create an array of Student structures and sort them based on different fields.

```
#include <stdio.h>
#include <string.h>

struct Student
{
   int id;
   char name[30];
   float grade;
};

int main()
{
   struct Student s[5], temp;
   int i, j, choice;

   for(i = 0; i < 5; i++)
   {
      printf("\nEnter details of student %d:\n", i + 1);

      printf("ID: ");
      scanf("%d", &s[i].id);

      printf("Name: ");
      scanf("%s", s[i].name);

      printf("Grade: ");
      scanf("%f", &s[i].grade);
   }

   printf("\nSort by:\n");
   printf("1. ID\n");
   printf("2. Name\n");
   printf("3. Grade\n");
   printf("Enter choice: ");
   scanf("%d", &choice);

   for(i = 0; i < 4; i++)
   {
      for(j = i + 1; j < 5; j++)
```

```
        {
            int condition = 0;

            if(choice == 1 && s[i].id > s[j].id)
                condition = 1;

            else if(choice == 2 && strcmp(s[i].name, s[j].name) > 0)
                condition = 1;

            else if(choice == 3 && s[i].grade > s[j].grade)
                condition = 1;

            if(condition)
            {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    printf("\nSorted Students:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d %s %.2f\n",
               s[i].id, s[i].name, s[i].grade);
    }

    return 0;
}
```

Output :

```
Enter details of student 1:
ID: 2025115020
Name: sripriyan
Grade: 80
```

```
Enter details of student 2:
ID: 22
Name: ace
Grade: 45
```

```
Enter details of student 3:
ID: 50
Name: luffy
Grade: 10

Enter details of student 4:
ID: 30
Name: zoro
Grade: 12

Enter details of student 5:
ID: 60
Name: sabo
Grade: 80

Sort by:
1. ID
2. Name
3. Grade
Enter choice: 2

Sorted Students:
22 ace 45.00
50 luffy 10.00
60 sabo 80.00
20 sripriyan 80.00
30 zoro 12.00

=== Code Execution Successful ===
```

9. Sort a set of negative numbers using radix sort.

```
#include <stdio.h>

int getMax(int arr[], int n)
{
    int max = arr[0];
    int i;

    for(i = 1; i < n; i++)
    {
        if(arr[i] > max)
            max = arr[i];
```

```
    }

    return max;
}

void radixPositive(int arr[], int n)
{
    int max = getMax(arr, n);
    int exp = 1;
    int output[100];
    int count[10];
    int i;

    while(max / exp > 0)
    {
        for(i = 0; i < 10; i++)
            count[i] = 0;

        for(i = 0; i < n; i++)
            count[(arr[i] / exp) % 10]++;

        for(i = 1; i < 10; i++)
            count[i] += count[i - 1];

        for(i = n - 1; i >= 0; i--)
        {
            output[count[(arr[i] / exp) % 10] - 1] = arr[i];
            count[(arr[i] / exp) % 10]--;
        }

        for(i = 0; i < n; i++)
            arr[i] = output[i];

        exp *= 10;
    }
}

int main()
{
    int arr[8] = {-45, 23, -12, 5, -8, 34, -30, 10};
    int pos[8], neg[8];
    int np = 0, nn = 0;
    int i, k = 0;
```

```
    for(i = 0; i < 8; i++)
    {
        if(arr[i] < 0)
            neg[nn++] = -arr[i];
        else
            pos[np++] = arr[i];
    }

    if(nn > 0)
        radixPositive(neg, nn);

    if(np > 0)
        radixPositive(pos, np);

    for(i = nn - 1; i >= 0; i--)
        arr[k++] = -neg[i];

    for(i = 0; i < np; i++)
        arr[k++] = pos[i];

    printf("Sorted array:\n");

    for(i = 0; i < 8; i++)
        printf("%d ", arr[i]);

    return 0;
}
```

Output :

```
Sorted array:
-45 -30 -12 -8 5 10 23 34

=== Code Execution Successful ===
```

10. Write a C program to copy one array to another using pointers.

```
#include <stdio.h>

int main()
{
```

```
    int a[5] = {10, 20, 30, 40, 50};
    int b[5];

    int *p = a;
    int *q = b;
    int i;

    for(i = 0; i < 5; i++)
    {
        *(q + i) = *(p + i);
    }

    printf("Copied array:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%d ", *(q + i));
    }

    return 0;
}
```

Output :

```
Copied array:
10 20 30 40 50

=== Code Execution Successful ===
```